

FitTrack AI

Personalized Fitness Recommendations Powered by AI

Phase 6 – Project Testing

Project type: AI-powered fitness tracking and recommendation REST API / backend

Stack: Node.js · Express.js · MongoDB Atlas · Mongoose · JWT · bcrypt.js · Google Gemini AI · REST APIs · Thunder Client / Postman

1. Test Plan

1.1 Testing Objective

To verify that the FitTrack AI REST API works as designed: users can register and log in securely, protected routes reject unauthorized requests, workouts can be created, read, updated, deleted and searched, the AI endpoints produce recommendations and insights through Gemini AI, invalid input is rejected, and every error is returned as a consistent JSON response.

1.2 Testing Scope

In scope: all implemented endpoints: `/api/auth/register`, `/api/auth/login`, `/api/auth/profile`, `/api/workouts` (create, list, get by ID, update, delete, search) and `/api/ai/recommendation`, `/api/ai/insights`, plus validation, authentication and error handling behavior.

Out of scope: front-end testing (the project has no front end), load and performance testing, and testing the internal quality of Google Gemini's generated text.

1.3 Testing Environment

- Local Node.js / Express.js server started with the development script
- MongoDB Atlas database, accessed through Mongoose (a test user and test workouts are used, not real personal data)
- Gemini API key and JWT secret provided through the `.env` file
- Thunder Client (VS Code) or Postman as the test client

1.4 Testing Tool

Thunder Client / Postman. Requests are saved in a collection with folders for Auth, Workouts, AI and Errors. The JWT returned by the login request is stored in an

environment variable and sent as `Authorization: Bearer <token>` on protected requests.

1.5 Testing Approach

The API is tested with manual black-box testing: each request is sent with known input, and the HTTP status code and JSON response are compared with the expected result. Each case is recorded as Pass or Fail. The recommended execution order is:

1. Authentication (a valid user and token are needed for everything else)
2. Workout CRUD and search
3. AI endpoints
4. Error and validation cases

Each test type is described below.

- **Functional API testing:** checks that each endpoint performs its intended function with valid input.
- **Authentication testing:** checks registration, login, JWT issuing, and rejection of missing or invalid tokens.
- **CRUD testing:** checks that workouts can be created, read, updated and deleted.
- **Search testing:** checks the search endpoint with the `q` parameter and without it.
- **AI integration testing:** checks that valid input reaches Gemini AI and that missing input or AI failure is handled properly. Only the API behavior is checked, not the wording of the AI output.
- **Validation testing:** checks that required fields, categories, durations and IDs are validated.
- **Error handling testing:** checks that all errors return the correct status and a JSON error response.

1.6 Test Plan Summary

2. API Test Cases

This table gives endpoint-level coverage of the whole API. Detailed test tables for each module follow in Sections 3 to 6.

3. Authentication Testing

Expected behavior: registration creates a user and never returns the plain-text password; duplicate emails and incomplete data are rejected; login returns a JWT only for correct credentials; the profile endpoint answers only when a valid JWT is supplied.

4. Workout Testing

Expected behavior: all workout routes require authentication. Valid data creates or updates a workout, invalid data is rejected with 400, a missing workout returns 404, and search returns matching workouts for the `q` parameter.

5. AI Testing

AI tests check the API behavior only. Gemini output varies from request to request, so the expected result is that a recommendation or insight is returned in the standard JSON format. Include the JWT in the request if the AI routes are protected in your implementation.

A. Workout Recommendation - `POST /api/ai/recommendation`

Input fields: `age`, `fitnessGoal`, `experience`. A valid request asks Gemini AI for a recommendation, missing input returns a validation error, and an AI or service failure is passed to the centralized error handler.

B. Fitness Insights - `POST /api/ai/insights`

Input fields: `totalWorkouts`, `averageDuration`, `totalCaloriesBurned`.

6. Error and Validation Testing

6.1 Centralized Error Handling

Controllers and services do not build error responses individually. When something fails (validation error, duplicate email, invalid token, invalid ID, database error, AI failure), the error is passed to `next(error)`, and a single error-handling middleware converts it into a JSON response with the right HTTP status code and message. Requests to unknown routes receive a 404 JSON response. This gives every failure

the same format and prevents stack traces and secrets from being returned to the client.

6.2 Error Test Cases

Test Result Summary

The *Total Test Cases* column counts the planned cases in the detailed tables of Sections 3 to 6. The other columns must stay as placeholders until the tests are executed. After running the test cases in Thunder Client / Postman, update each Status in the tables above, count them, and enter the final values below.

Testing Screenshot Evidence

Capture these screenshots from the working project in Thunder Client / Postman and insert them in the report. Each should show the request (method, URL, body), the response status code and the JSON response. Do not use mock-ups. **Hide or blur the JWT, database credentials and API keys before adding any screenshot to the report.**

Testing Flow Diagram

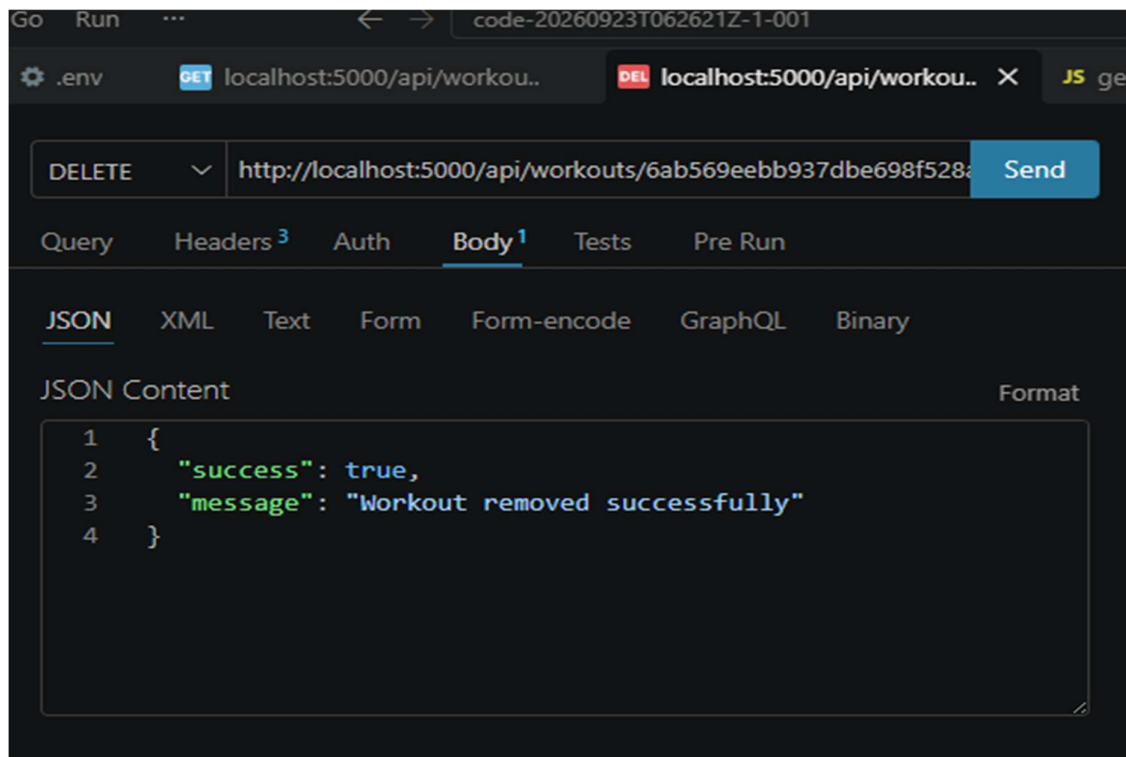
FitTrack AI – API Testing Flow Thunder Client / Postman
Tester prepares method, URL, headers, body API Request
JSON body + Bearer JWT (protected routes)
Express Server JSON parsing · CORS · logging Route /api/auth · /api/workouts · /api/ai (unknown route → 404)
Authentication Middleware Verify JWT on protected routes
Controller Validate input · call service Service / Model Business logic · Mongoose models MongoDB Atlas Users and Workouts
Gemini AI Recommendation / Insight API Response HTTP status code + JSON body
Test Result Compare expected vs actual → Pass / Fail 401 / errors → centralized error handler → JSON error response
Fail → fix the issue and re-run Legend Request Server Authentication Business logic Database AI Response
Figure 1: FitTrack AI – API Testing Flow (dashed red: error path; dashed grey: re-test after a failed result)

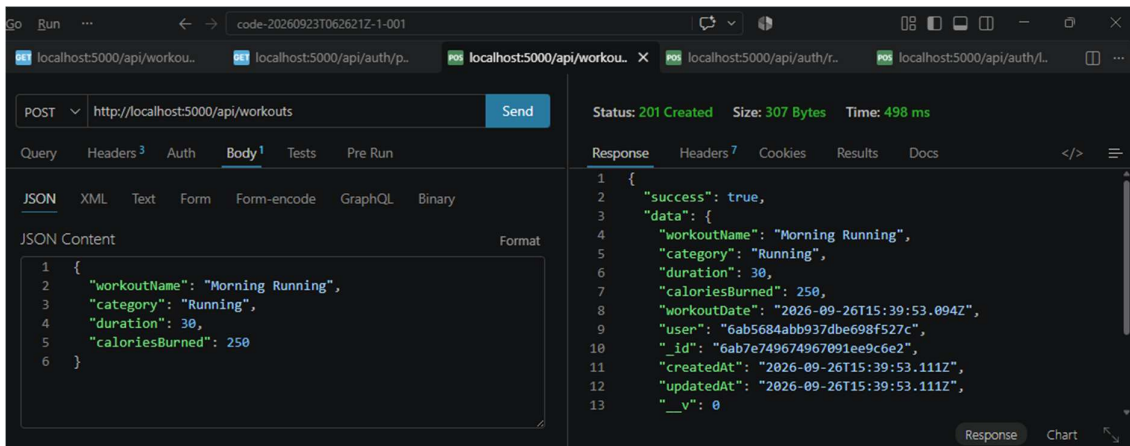
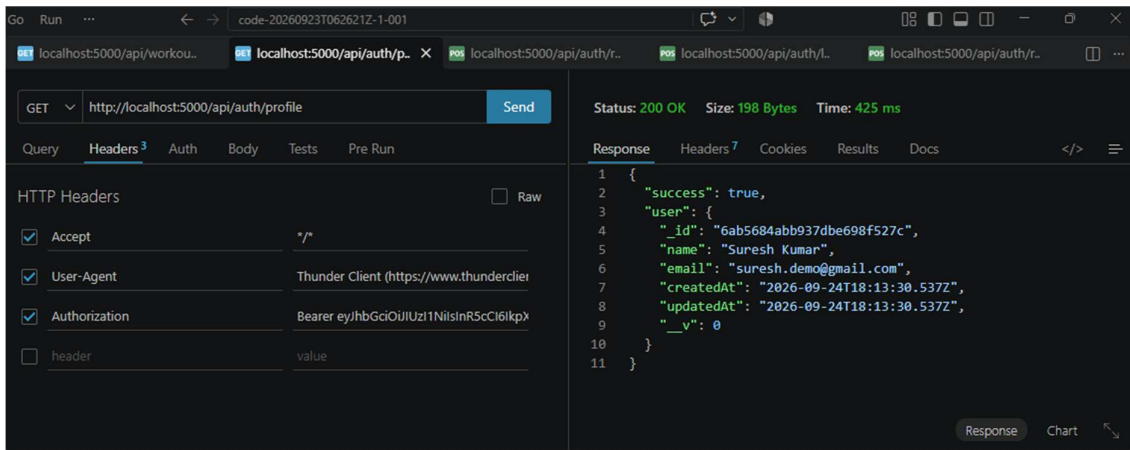
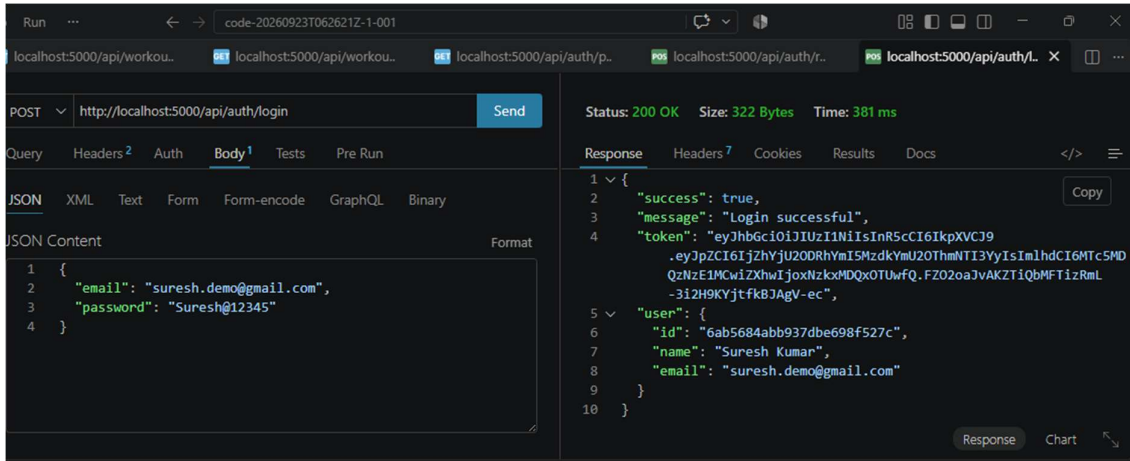
Phase 6 Conclusion

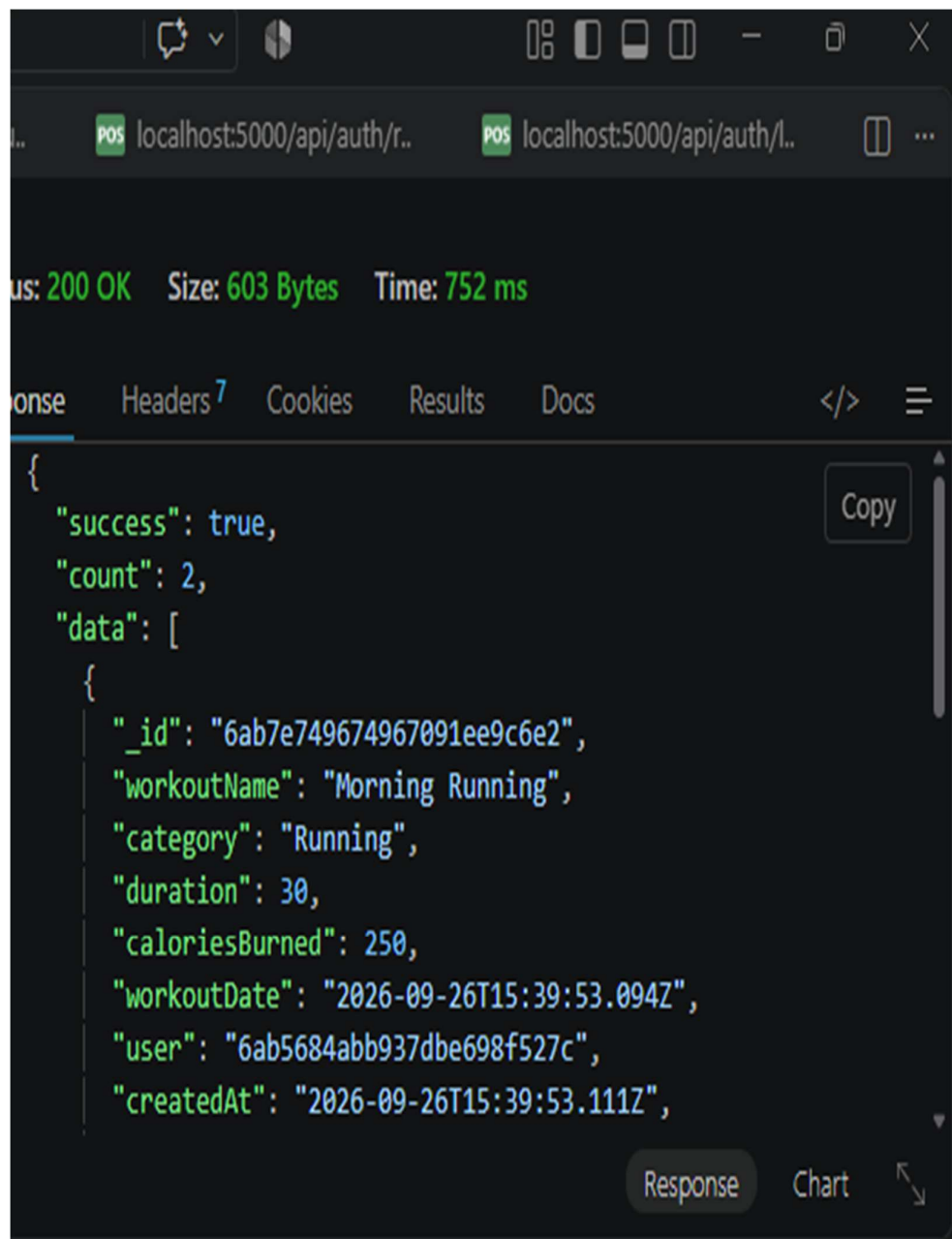
Testing is used to verify that FitTrack AI works reliably and safely before it is presented. The test cases in this phase cover:

- **API functionality:** each endpoint behaves as designed
- **Authentication:** registration, login, JWT-protected profile and rejection of missing or invalid tokens
- **Workout CRUD:** creating, reading, updating and deleting workouts
- **Search:** finding workouts with the `q` parameter
- **AI features:** recommendation and fitness insight requests through Gemini AI
- **Input validation:** missing, invalid or unsupported data is rejected
- **Error handling:** centralized handling with correct status codes
- **JSON responses:** a consistent response format for both success and errors

SCREENSHOTS FOR EXECUTION







localhost:5000/api/workou... localhost:5000/api/auth/p... localhost:5000/api/ai/fit... X localhost:5000/api/workou... localhost:5000/api/workou... [] ...

POST http://localhost:5000/api/ai/fitness-insights Send

Query Headers³ Auth Body¹ Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
1 {
2   "totalWorkouts": 10,
3   "averageDuration": 45,
4   "totalCaloriesBurned": 3200
5 }
```

Status: 200 OK Size: 415 Bytes Time: 6.61 s

Response Headers⁷ Cookies Results Docs </> ≡

```
1 {
2   "insight": "You have built strong momentum with 10 consistent 45
-minute workouts, burning an impressive total of 3,200
calories. To take your progress to the next level, try
adding five minutes to your sessions or slightly increasing
your training intensity to keep challenging your
cardiovascular system. Lock in this habit by scheduling your
next three workouts on your calendar today to maintain your
momentum."
3 }
```

Response Chart ↗

Status: 200 OK Size: 86 Bytes Time: 10 ms

Response Headers⁷ Cookies Results Docs </> ≡

```
1 {
2   "success": true,
3   "message": "AI FitTrack API is running successfully",
4   "version": "1.0.0"
5 }
```

Copy

Response Chart ↗